Solutions to 2010 Power Question

1a. By symmetry, P_1, P_2, the two plots sold on day 1, are centered on the y-axis, say at $(0, \pm y)$ with $y > 0$. Let these plots have radius r. Because P_1 is tangent to U, $y + r = 1$. Because P_1 is tangent to C, the distance from $(0, 1-r)$ to $\left(\frac{1}{2}, 0\right)$ is $r + \frac{1}{2}$. Therefore

$$\begin{aligned} \left(\frac{1}{2}\right)^2 + (1-r)^2 &= \left(r + \frac{1}{2}\right)^2 \\ 1 - 2r &= r \\ r &= \frac{1}{3} \\ y = 1 - r &= \frac{2}{3}. \end{aligned}$$

Thus the plots are centered at $\left(0, \pm\frac{2}{3}\right)$ and have radius $\frac{1}{3}$.

1b. The four "removed" circles have radii $\frac{1}{2}, \frac{1}{2}, \frac{1}{3}, \frac{1}{3}$ so the combined area of the six remaining curvilinear territories is:

$$\pi\left(1^2 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{3}\right)^2 - \left(\frac{1}{3}\right)^2\right) = \frac{5\pi}{18}.$$

2a. At the beginning of day 2, there are six c-triangles, so six incircles are sold, dividing each of the six territories into three smaller curvilinear triangles. So a total of 18 curvilinear triangles exist at the start of day 3, each of which is itself divided into three pieces that day (by the sale of a total of 18 regions bounded by the territories' incircles). Therefore there are 54 regions at the end of day 3.

2b. Each day, every curvilinear territory is divided into three smaller curvilinear territories. Let R_n be the number of regions at the end of day n. Then $R_0 = 2$, and $R_{n+1} = 3 \cdot R_n$. Thus R_n is a geometric sequence, so $R_n = 2 \cdot 3^n$. For $n > 0$, the number of plots sold on day n equals the number of territories existing at the end of day $n-1$, i.e., $X_n = R_{n-1}$, so $X_0 = X_1 = 2$, and for $n > 1, X_n = 2 \cdot 3^{n-1}$.

2c. The total number of plots sold up to and including day n is

$$\begin{aligned} 2 + \sum_{k=1}^{n} X_k &= 2 + 2\sum_{k=1}^{n} 3^{k-1} \\ &= 2 + 2 \cdot (1 + 3 + 3^2 + \ldots + 3^{n-1}) \\ &= 3^n + 1. \end{aligned}$$

Alternatively, proceed by induction: on day 0, there are $2 = 3^0 + 1$ plots sold, and for $n \geq 0$,

$$\begin{aligned} (3^n + 1) + X_{n+1} &= (3^n + 1) + 2 \cdot 3^n \\ &= 3 \cdot 3^n + 1 \\ &= 3^{n+1} + 1. \end{aligned}$$

3a. Use Descartes' Circle Formula with $a = b = 1$ and $c = \frac{3}{2}$ to solve for d:

$$\begin{aligned} 2 \cdot \left(1^2 + 1^2 + \left(\frac{3}{2}\right)^2 + d^2\right) &= \left(1 + 1 + \frac{3}{2} + d\right)^2 \\ \frac{17}{2} + 2d^2 &= \frac{49}{4} + 7d + d^2 \\ d^2 - 7d - \frac{15}{4} &= 0, \end{aligned}$$

from which $d = \frac{15}{2}$ or $d = -\frac{1}{2}$. These values correspond to radii of $\frac{2}{15}$, a small circle nestled between the other three, or 2, a large circle enclosing the other three.

Alternatively, start by scaling the kingdom with the first four circles removed to match the situation given. Thus the three given circles are internally tangent to a circle of radius $r = 2$ and curvature $d = -\frac{1}{2}$. Descartes' Circle Formula gives a quadratic equation for d, and the sum of the roots is $2 \cdot (1 + 1 + \frac{3}{2}) = 7$, so the second root is $7 + \frac{1}{2} = \frac{15}{2}$, corresponding to a circle of radius $r = \frac{2}{15}$.

3b. Apply Descartes' Circle Formula to yield

$$(a + b + c + x)^2 = 2 \cdot (a^2 + b^2 + c^2 + x^2),$$

a quadratic equation in x. Expanding and rewriting in standard form yields the equation

$$x^2 - px + q = 0$$

where $p = 2(a + b + c)$ and $q = 2(a^2 + b^2 + c^2) - (a + b + c)^2$.

The discriminant of this quadratic is

$$\begin{aligned} p^2 - 4q &= 8(a + b + c)^2 - 8(a^2 + b^2 + c^2) \\ &= 16(ab + ac + bc). \end{aligned}$$

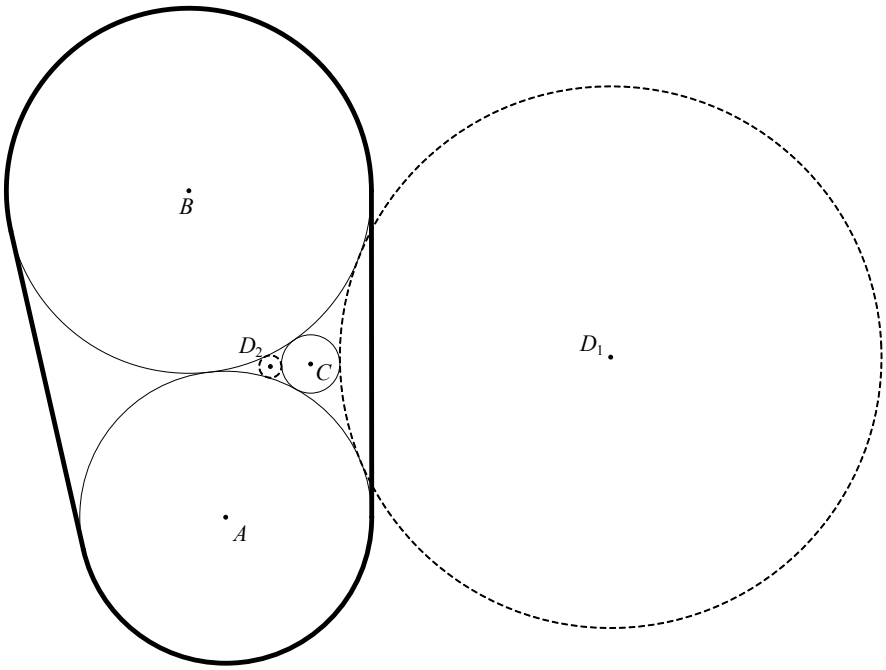
This last expression is positive because it is given that $a, b, c > 0$. Therefore the quadratic has two distinct real roots, say d_1 and d_2. These usually correspond to two distinct radii, $r_1 = \dfrac{1}{|d_1|}$ and $r_2 = \dfrac{1}{|d_2|}$.

There are two possible exceptions. If $(a, b, c, 0)$ satisfies Descartes' Circle Formula, then one of the radii is undefined. The other case to consider is if $r_2 = r_1$, which would occur if $d_2 = -d_1$. This case can be ruled out because $d_1 + d_2 = p = 2(a + b + c)$, which must be positive if $a, b, c > 0$. (Notice too that this inequality rules out the possibility that both circles have negative curvature, so that there cannot be two distinct circles to which the given circles are internally tangent.)

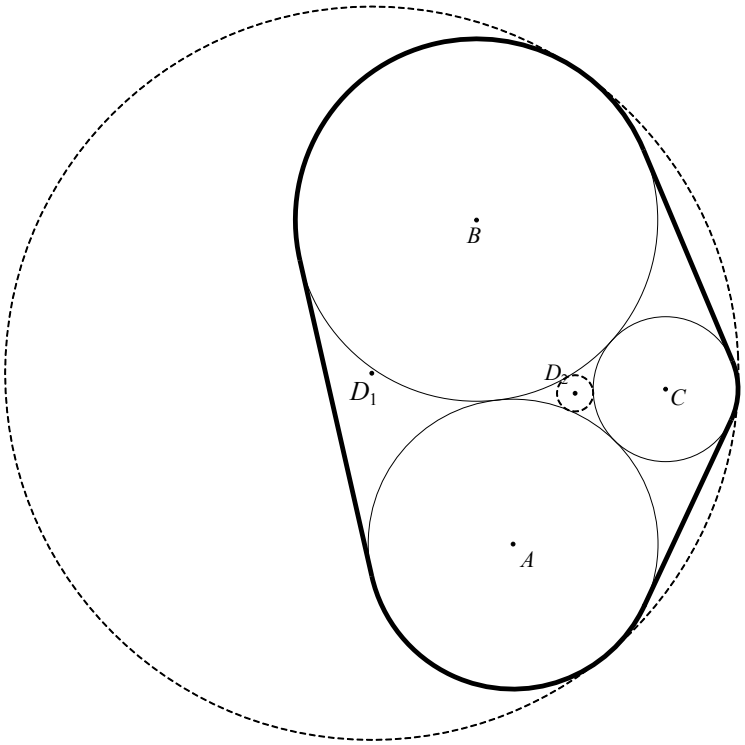
When both roots d_1 and d_2 are positive, the three given circles are externally tangent to both fourth circles. When one is positive and one is negative, the three given circles are internally tangent to one circle and externally tangent to the other.

While the foregoing answers the question posed, it is interesting to examine the result from a geometric perspective: why are there normally two possible fourth circles? Consider the case when one of a, b, and c is negative (i.e., two circles are internally tangent to a third). Let A and B be circles internally tangent to C. Then A and B partition the remaining area of C into two c-triangles, each of which has an incircle, providing the two solutions.

If, as in the given problem, $a, b, c > 0$, then all three circles A, B, and C, are mutually externally tangent. In this case, the given circles bound a c-triangle, which has an incircle, corresponding to one of the two roots. The complementary arcs of the given circles bound an infinite region, and this region normally contains a second circle tangent to the given circles. To demonstrate this fact geometrically, consider shrink-wrapping the circles: the shrink-wrap is the border of the smallest convex region containing all three circles. (This region is called the *convex hull* of the circles). There are two cases to address. If only two circles are touched by the shrink-wrap, then one circle is wedged between two larger ones and completely enclosed by their common tangents. In such a case, a circle can be drawn so that it is tangent to all three circles as shown in the diagram below (shrink-wrap in bold; locations of fourth circle marked at D_1 and D_2).



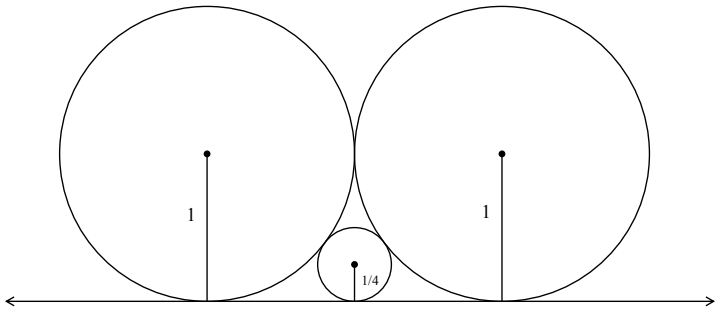
On the other hand, if the shrink-wrap touches all three circles, then it can be expanded to make a circle tangent to and containing A, B, and C, as shown below.



The degenerate case where $(a, b, c, 0)$ satisfies Descartes' Circle Formula is treated in 3c below.

One final question is left for the reader to investigate. Algebraically, it is possible that there is a double root if $p^2 - 4q = 0$. To what geometric situation does this correspond, and under what (geometric) conditions can it arise?

3c. In this case, the fourth "circle" is actually a line tangent to all three circles, as shown in the diagram below.



4a. Note that

$$\begin{aligned} \frac{1}{\rho} &= \frac{\phi^2 - \phi}{\phi + \sqrt{\phi}} \\ &= \phi - \sqrt{\phi}\,. \end{aligned}$$

Therefore

$$\begin{aligned}\left(\rho - \frac{1}{\rho}\right)^2 &= \left(2\sqrt{\phi}\right)^2 = 4\phi \\ &= 2\left(\rho + \frac{1}{\rho}\right).\end{aligned}$$

Multiplying both sides of the equation by ρ^2 gives $(\rho^2-1)^2 = 2(\rho^3+\rho)$. Expand and isolate ρ^4 to obtain $\rho^4 = 2\rho^3 + 2\rho^2 + 2\rho - 1$.

Alternate Proof: Because $\phi^2 = \phi + 1$, any power of ρ can be expressed as an integer plus integer multiples of $\sqrt{\phi}$, ϕ, and $\phi\sqrt{\phi}$. In particular,

$$\begin{aligned}\rho^2 &= \phi^2 + 2\phi\sqrt{\phi} + \phi \\ &= 2\phi\sqrt{\phi} + 2\phi + 1, \\ \rho^3 &= 4\phi\sqrt{\phi} + 5\phi + 3\sqrt{\phi} + 4, \text{ and} \\ \rho^4 &= 12\phi\sqrt{\phi} + 16\phi + 8\sqrt{\phi} + 9.\end{aligned}$$

Therefore

$$\begin{aligned}2\rho^3 + 2\rho^2 + 2\rho - 1 &= 2\left(4\phi\sqrt{\phi} + 5\phi + 3\sqrt{\phi} + 4\right) + 2\left(2\phi\sqrt{\phi} + 2\phi + 1\right) + 2\rho - 1 \\ &= 12\phi\sqrt{\phi} + 16\phi + 8\sqrt{\phi} + 9 \\ &= \rho^4.\end{aligned}$$

4b. If the radii are in geometric progression, then so are their reciprocals (i.e., curvatures). Without loss of generality, let $(a,b,c,d) = (a, ar, ar^2, ar^3)$ for $r > 1$. By Descartes' Circle Formula,

$$\left(a + ar + ar^2 + ar^3\right)^2 = 2\left(a^2 + a^2r^2 + a^2r^4 + a^2r^6\right).$$

Cancel a^2 from both sides of the equation to obtain

$$\left(1 + r + r^2 + r^3\right)^2 = 2\left(1 + r^2 + r^4 + r^6\right).$$

Because $1 + r + r^2 + r^3 = (1+r)(1+r^2)$ and $1 + r^2 + r^4 + r^6 = (1+r^2)(1+r^4)$, the equation can be rewritten as follows:

$$\begin{aligned}(1+r)^2\left(1+r^2\right)^2 &= 2\left(1+r^2\right)\left(1+r^4\right) \\ (1+r)^2\left(1+r^2\right) &= 2\left(1+r^4\right) \\ r^4 - 2r^3 - 2r^2 - 2r + 1 &= 0.\end{aligned}$$

Using the identity from 4a, $r = \rho$ is one solution; because the polynomial is palindromic, another real solution is $r = \rho^{-1} = \phi - \sqrt{\phi}$, but this value is less than 1. The product of the corresponding linear factors is $r^2 - 2\phi r + \phi^2 - \phi = r^2 - 2\phi r + 1$. Division verifies that the other quadratic factor of the polynomial is $r^2 + (2\phi - 2)r + 1$, which has no real roots because $(\sqrt{5}-1)^2 < 1$.

5. The equation $(x+b+c+d)^2 = 2(x^2+b^2+c^2+d^2)$ is quadratic with two solutions. Call them a and a'. These are the curvatures of the two circles which are tangent to circles with curvatures b, c, and d. Rewrite the equation in standard form to obtain $x^2 - 2(b+c+d)x + \ldots = 0$. Using the sum of the roots formula, $a + a' = 2(b+c+d) = 2(s-a)$. So $a' = 2s - 3a$, and therefore

$$\begin{aligned} s' &= a' + b + c + d \\ &= 2s - 3a + s - a \\ &= 3s - 4a. \end{aligned}$$

6. Day 1 starts with circles of curvature $-1, 2, 2$ bounding C and C'. The geometers mark off P, P' with curvature 3 yielding configurations $(-1, 2, 2, 3)$, $s = 6$. Then the king sells two plots of curvature 3.

   a. To find plots sold on day 2 start with the configuration $(-1, 2, 2, 3)$ and compute the three distinct curvatures of circles conjugate to one of the circles in this configuration. Because $s = 6$, the curvatures are $2 \cdot 6 - 3(-1) = 15$, $2 \cdot 6 - 3 \cdot 2 = 6$, and $2 \cdot 6 - 3 \cdot 3 = 3$. However, if P is the circle of curvature 3 included in the orientation, then P' is the new conjugate circle of curvature 3, which was also marked off on day 2. Thus the only options are 15 and 6. In the first case, $s' = 3s - 4a = 3 \cdot 6 - 4(-1) = 22$; in the second, $s' = 3s - 4a = 3 \cdot 6 - 4 \cdot 2 = 10$.

   b. On day 2, six plots are sold: two with curvature 15 from the configuration $(2, 2, 3, 15)$, and four with curvature 6 from the configuration $(-1, 2, 3, 6)$. The total area sold on day 2 is therefore
   $$2 \cdot \frac{\pi}{15^2} + 4 \cdot \frac{\pi}{6^2} = \frac{3}{25}\pi,$$
   which is exactly 12% of the unit circle.

   c. Day 3 begins with two circles of curvature 15 from the configuration $(2, 2, 3, 15)$, and four circles of curvature 6 from the configuration $(-1, 2, 3, 6)$. Consider the following two cases:

Case 1: $(a, b, c, d) = (2, 2, 3, 15)$, $s = 22$

- $a = 2 : a' = 2s - 3a = \mathbf{38}$
- $b = 2 : b' = 2s - 3b = \mathbf{38}$
- $c = 3 : c' = 2s - 3c = \mathbf{35}$
- $d = 15 : d' = 2s - 3d = -1$, which is the configuration from day 1.

Case 2: $(a, b, c, d) = (-1, 2, 3, 6)$, $s = 10$

- $a = -1 : a' = 2s - 3a = \mathbf{23}$
- $b = 2 \ : b' = 2s - 3b = \mathbf{14}$
- $c = 3 \ : c' = 2s - 3c = \mathbf{11}$
- $d = 6 \ : d' = 2s - 3d = 2$, which is the configuration from day 1.

So the areas of the plots removed on day 3 are:

$$\frac{\pi}{38^2},\ \frac{\pi}{35^2},\ \frac{\pi}{23^2},\ \frac{\pi}{14^2},\ \text{and}\ \frac{\pi}{11^2}.$$

There are two circles with area $\frac{\pi}{35^2}$, and four circles with each of the other areas, for a total of 18 plots.

d. Because 18 plots were sold on day 3, the mean curvature is

$$\frac{2(38+38+35)+4(23+14+11)}{18} = 23.$$

7. Proceed by induction. The base case, that all curvatures prior to day 2 are integers, was shown in problem 1a. Using the formula $a' = 2s - 3a$, if a, b, c, d, and s are integers on day n, then a', b', c', and d' are integer curvatures on day $n+1$, proving inductively that all curvatures are integers.

8a. Notice that substituting az_A, bz_B, cz_C, dz_D for a, b, c, d respectively in the derivation of the formula in problem 4 leaves the algebra unchanged, so in general, $a'z_{A'} = 2\hat{s} - 3az_A$, and similarly for $b'z_{B'}, c'z_{C'}, d'z_{D'}$.

8b. It suffices to show that for each circle C with curvature c and center z_C (in the complex plane), cz_C is of the form $u + iv$ where u and v are integers. If this is the case, then each center is of the form $\left(\frac{u}{c}, \frac{v}{c}\right)$.

Proceed by induction. To check the base case, check the original kingdom and the first four plots: U, C, C', P_1, and P_2. Circle U is centered at $(0,0)$, yielding $-1 \cdot z_U = 0 + 0i$. Circles C and C' are symmetric about the y-axis, so it suffices to check just one of them. Circle C has radius $\frac{1}{2}$ and therefore curvature 2. It is centered at $\left(\frac{1}{2}, \frac{0}{2}\right)$, yielding $2z_C = 2\left(\frac{1}{2} + 0i\right) = 1$. Circles P_1 and P_2 are symmetric about the x-axis, so it suffices to check just one of them. Circle P_1 has radius $\frac{1}{3}$ and therefore curvature 3. It is centered at $\left(\frac{0}{3}, \frac{2}{3}\right)$, yielding $3z_{P_3} = 0 + 2i$.

For the inductive step, suppose that $a\,z_A, b\,z_B, c\,z_C, d\,z_D$ have integer real and imaginary parts. Then by closure of addition and multiplication in the integers, $a'z_{A'} = 2\hat{s} - 3a\,z_A$ also has integer real and imaginary parts, and similarly for $b'z_{B'}, c'z_{C'}, d'z_{D'}$.

So for all plots A sold, $a\,z_A$ has integer real and imaginary parts, so each is centered at $\left(\frac{u}{c}, \frac{v}{c}\right)$ where u and v are integers, and c is the curvature.

9a. Let $s = a + b + c + d$. From problem 5, it follows that

$$\begin{aligned}
\mathcal{C}(P^{(1)}) &= (b, c, d, a'), \\
\mathcal{C}(P^{(2)}) &= (a, c, d, b'), \\
\mathcal{C}(P^{(3)}) &= (a, b, d, c'),
\end{aligned}$$

where $a' = 2s - 3a$, $b' = 2s - 3b$, and $c' = 2s - 3c$. Because $a \le b \le c$, it follows that $c' \le b' \le a'$. Therefore $P^{(3)}$ dominates $P^{(2)}$ and $P^{(2)}$ dominates $P^{(1)}$.

9b. Because $\mathcal{C}(P^{(1)}) = (b, c, d, a')$ and $\mathcal{C}(Q^{(1)}) = (x, y, z, w')$, it is enough to show that $a' \le w'$. As in the solution to problem 5, a and a' are the two roots of the quadratic given by Descartes' Circle Formula:

$$(X + b + c + d)^2 = 2(X^2 + b^2 + c^2 + d^2).$$

Solve by completing the square:

$$\begin{aligned} X^2 - 2(b + c + d)X + 2(b^2 + c^2 + d^2) &= (b + c + d)^2; \\ \big(X - (b + c + d)\big)^2 &= 2(b + c + d)^2 - 2(b^2 + c^2 + d^2) \\ &= 4(bc + bd + cd). \end{aligned}$$

Thus a, $a' = b + c + d \pm 2\sqrt{bc + bd + cd}$.

Because $a \le b \le c \le d$, and only a can be less than zero, a must get the minus sign, and a' gets the plus sign:

$$a' = b + c + d + 2\sqrt{bc + bd + cd}.$$

Similarly,

$$w' = x + y + z + 2\sqrt{xy + xz + yz}.$$

Because P dominates Q, each term in the expression for a' is less than or equal to the corresponding term in the expression for w', thus $a' \le w'$.

9c. Because $\mathcal{C}(P^{(3)}) = (a, b, d, c')$ and $\mathcal{C}(Q^{(3)}) = (w, x, z, y')$, it suffices to show that $c' \le y'$. If $a \ge 0$, then the argument is exactly the same as in problem 9a, but if $a < 0$, then there is more to be done.

Arguing as in 9b, c, $c' = a + b + d \pm 2\sqrt{ab + ad + bd}$. If $a < 0$, then the other three circles are *internally* tangent to the circle of curvature a, so this circle has the largest radius. In particular, $\frac{1}{|a|} > \frac{1}{b}$. Thus $b > |a| = -a$, which shows that $a + b > 0$. Therefore c must get the minus sign, and c' gets the plus sign. The same argument applies to y and y'.

When $a < 0$, it is also worth considering whether the square roots are defined (and real). In fact, they are. Consider the diameters of the circles with curvatures b and d along the line through the centers of these circles. These two diameters form a single segment inside the circle with curvature a, so the sum of the diameters is at most the diameter of that circle: $\frac{2}{b} + \frac{2}{d} \le \frac{2}{|a|}$. It follows that $-ad - ab = |a|\,d + |a|\,b \le bd$, or $ab + ad + bd \ge 0$. This is the argument of the square root in the expressions for c and c'. An analogous argument shows that the radicands are nonnegative in the expressions for b and b'.

The foregoing shows that

$$c' = a + b + d + 2\sqrt{ab + ad + bd}$$

and, by an analogous argument for $w < 0$,

$$y' = w + x + z + 2\sqrt{wx + wz + xz}.$$

It remains to prove that $c' \le y'$. Note that only a and w may be negative; b, c, d, x, y, and z are all positive. There are three cases.

(**i**) If $0 \le a \le w$, then $ab \le wx$, $ad \le wz$, and $bd \le xz$, so $c' \le y'$.

(**ii**) If $a < 0 \le w$, then $ab + ad + bd \le bd$, and $bd \le xz \le wx + wz + xz$, so $c' \le y'$. (As noted above, both radicands are nonnegative.)

(**iii**) If $a \le w < 0$, then it has already been established that $a + b$ is positive. Analogously, $a + d$, $w + x$, and $w + z$ are positive. Furthermore, $a^2 \ge w^2$. Thus $(a + b)(a + d) - a^2 \le (w + x)(w + z) - w^2$, which establishes that $ab + ad + bd \le wx + wz + xz$, so $c' \le y'$.

10a. First, show by induction that every c-triangle on every day in the kingdom is proper. For the base case, both c-triangles at the end of day 0 have configuration $(-1, 2, 2, 3)$, so they are proper. For the inductive step, let T be a proper c-triangle. If $\mathcal{C}(T) = (a, b, c, d)$, then the three c-triangles obtained from T on the next day have configurations $\mathcal{C}(T^{(1)}) = (b, c, d, a')$, $\mathcal{C}(T^{(2)}) = (a, c, d, b')$, and $\mathcal{C}(T^{(3)}) = (a, b, d, c')$. According to problem 5, $c' = 2a+2b-c+2d = 2(a + b) + (d - c) + d$. Arguing as in the proof of 9c, $a + b \ge 0$. By the inductive hypothesis, T is proper, $d - c \ge 0$; therefore $c' \ge d$. Because $a' \ge b' \ge c'$, all three c-triangles are proper.

If $n = 0$, then both circles sold on day n have curvature 2, and this fits the formula: $n^2 + 2 = 0^2 + 2 = 2$.

Let P_0 be one of the c-triangles left at the end of day 0. For $m > 0$, let $P_m = P_{m-1}^{(3)}$. Use induction to prove the following two claims:

(i) $\mathcal{C}(P_m) = (-1, 2, m^2 + 2, (m + 1)^2 + 2)$.

(ii) P_m dominates all c-triangles left at the end of day m.

For the moment, grant these two claims. Then (ii) implies that the incircle of P_{n-1} is at least as large as any plot sold on day n, and (i) shows that this incircle has curvature $n^2 + 2$.

For the base case, both c-triangles at the end of day 0 have associated circle configuration $\mathcal{C}(P_0) = (-1, 2, 2, 3) = (-1, 2, 0^2 + 2, 1^2 + 2)$, so either dominates the other.

For $m > 0$, assume inductively that $\mathcal{C}(P_{m-1}) = (-1, 2, (m - 1)^2 + 2, m^2 + 2) = (a, b, c, d)$. Because $P_m = P_{m-1}^{(3)}$, $\mathcal{C}(P_m) = (a, b, d, c') = (-1, 2, m^2 + 2, c')$. Use algebra and the result of problem 5 to obtain $c' = 2(a + b + c + d) - 3c = (m + 1)^2 + 2$. This completes the inductive step for (i).

Now let Q be any c-triangle left at the end of day $m - 1$, with $\mathcal{C}(Q) = (x, y, z, w)$. Any c-triangle left at the end of day m is of the form $Q^{(1)}$, $Q^{(2)}$, or $Q^{(3)}$ for some such Q. By the inductive hypothesis, P_{m-1} dominates Q. It has already been established that these c-triangles are both proper, so the results of problem 9 apply. By 9c, $P_m = P_{m-1}^{(3)}$ dominates $Q^{(3)}$, and by 9a, $Q^{(3)}$ dominates $Q^{(1)}$ and $Q^{(2)}$. This completes the inductive step for (ii).

10b. Let R_n be a c-triangle with configuration

$$\mathcal{C}(R_n) = (2\rho^{n-2}, 2\rho^{n-1}, 2\rho^n, 2\rho^{n+1}).$$

According to problem 4, these four numbers (a geometric progression with common ratio ρ) satisfy Descartes' Circle Formula, so there is such a c-triangle. The following inductive argument proves that for $n \ge 2$, each c-triangular plot remaining in the kingdom at the end of day n dominates R_n.

For the base case, problem 6 shows that the following are sufficient: $2 \leq 2\rho^0$, $3 \leq 2\rho^1$, $15 \leq 2\rho^2$, and $38 \leq 2\rho^3$. In fact, it is enough to calculate $\phi = \frac{1+\sqrt{5}}{2} > 1.6$, $\sqrt{\phi} > 1.2$, $\rho = \phi + \sqrt{\phi} > 2.8$ to conclude that $2\rho > 5.6$, $2\rho^2 > 15.68$, and $2\rho^3 > 43.9$. These same calculations show that the main result is true for $n \leq 2$.

For the inductive step, let T be a c-triangle in the kingdom remaining at the end of day n. The inductive hypothesis is that T dominates R_n. By problems 9a and 9b, all the c-triangles obtained from T on day $n+1$ dominate $R_n^{(1)}$. All that remains is to show that $R_n^{(1)}$ has the right configuration, so that $R_{n+1} = R_n^{(1)}$.

One approach is to use the formulas from problems 4a and 5 to show that $2(2\rho^{n-2} + 2\rho^{n-1} + 2\rho^n + 2\rho^{n+1}) - 3 \cdot 2\rho^{n-2} = 2\rho^{n+2}$. A method that avoids calculation is to note that $\mathcal{C}(R_n^{(1)}) = (2\rho^{n-1}, 2\rho^n, 2\rho^{n+1}, x)$ for some x. Because Descartes' Circle Formula is quadratic in x, there are at most two possibilities. According to 4b, two solutions are given by $x = 2\rho^{n-2}$ and $x = 2\rho^{n+2}$, because both of these give geometric progressions with common ratio ρ. The first corresponds to R_n, so the second must correspond to ${R_n}^{(1)}$. This completes the induction.

If $n \leq 2$, then (as already noted) the solution to problem 6 shows that the curvature of the smallest plot sold on day n does not exceed $2\rho^n$. If $n > 2$, then this smallest plot is the incircle of some c-triangle T that remains at the end of day $n-1$, with $n-1 \geq 2$. Because T dominates R_{n-1}, the curvature of its incircle does not exceed that of R_{n-1}, which is $2\rho^n$.